SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I **D Vaishnavi (192524143)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

   Ans:

Given Constraints

- D1 cannot work **Night**.
- D2 must work **before** D3 (Morning < Afternoon < Night).
- Only **one doctor per shift**.
- D3 cannot work **Morning**.
- Every doctor is assigned **exactly one shift**.

Step 1: Available Domains

- D1 = {Morning, Afternoon}
- D2 = {Morning, Afternoon, Night}
- D3 = {Afternoon, Night}

Step 2: Backtracking Search

Try 1

- D1 → Morning
- D2 → Morning (Morning already assigned)

  Backtrack.

Try 2

- D2 → Afternoon
- D3 → Afternoon (Already assigned)

  Backtrack.

Try 3

- D3 → Night

Check constraints:

- D1 ≠ Night
- D2 before D3 (Afternoon before Night)
- One doctor per shift
- D3 ≠ Morning
- All doctors assigned once

Final Valid Schedule

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	Night

Answer

Morning → D1
Afternoon → D2
Night → D3

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

i. Movement allowed: Up, Down, Left, Right
ii. Cost of each move = 1
iii. Cannot pass through obstacles
iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Ans:

Given

- Grid = 5 × 5
- Moves = Up, Down, Left, Right
- Cost of each move = 1
- Obstacles cannot be crossed
- Manhattan Distance heuristic is given.

BFS Steps

1. Start from **S**.
2. Insert **S** into the queue.
3. Remove the first node.
4. Expand all valid neighbouring nodes.
5. Ignore obstacles and visited nodes.
6. Insert new nodes into the queue.
7. Repeat until **Goal (G)** is reached.
8. Trace the parent nodes to obtain the shortest path.

Queue Update

Initial Queue: `[S]`

Expand S: `[A, B]`

Expand A: `[B, C]`

Expand B: `[C, D]`

Expand C: `[D, E]`

Expand D: `[E, G]`

Goal reached.

Optimal Path

S → A → C → E → G

Cost Calculation

Each move = 1

S → A = 1

A → C = 1

C → E = 1

E → G = 1

Total Cost = 4

Answer

- Optimal Path: S → A → C → E → G
- Total Cost = **4**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Ans:

(a) Constraint Analysis

Constraints

1. Robot moves only Up, Down, Left and Right.
2. Obstacles cannot be crossed.
3. Robot senses only adjacent cells.
4. Normal movement cost = 1.
5. Risky zone adds extra cost = 2.
6. Robot updates knowledge dynamically.

Effect on Decision Making

- Limited movement reduces available paths.
- Obstacles force the robot to find alternate routes.
- Limited sensing means the robot knows only nearby cells.
- Risky zones increase total cost.
- Dynamic updates help the robot avoid newly discovered obstacles.

(b) Solution Approach (Online Search / Uniform Cost Search)

Steps

1. Start at S.
2. Observe neighbouring cells.
3. Ignore blocked cells.
4. Calculate movement cost.
5. If risky zone:

    - Cost = 1 + 2 = 3

6. Select the path with the minimum cumulative cost.
7. Move to the selected cell.
8. Update knowledge after every move.
9. Continue until Goal (G) is reached.

(c) AI Concepts

Intelligent Agent

The rescue robot is an intelligent agent because it senses the environment, processes information and performs actions to achieve the goal.

Rationality

The robot chooses the safest and least-cost path to maximize performance.

Environment Type

- Partially observable
- Dynamic
- Sequential
- Discrete
- Single-agent

(d) Justification

The **Online Search / Uniform Cost Search** approach is the most suitable because:

- It works well in partially observable environments.
- It always selects the minimum-cost path.
- It updates decisions dynamically.
- It avoids blocked paths.
- It minimizes total travel cost while ensuring safe navigation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided